

Mini Project Report - 08

Master of Computer Application – General
Semester – I

Sub: Front-End Frameworks & Technologies

Topic: JavaScript Grade Calculator

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INTRODUCTION

The **Grade Calculator** is a simple web-based application developed using **HTML, CSS, and JavaScript**. It allows students to enter marks for seven subjects: FFT, ADS, BAD, PST, CLS, AUI, and DST. The application validates the entered marks, calculates the **total marks and percentage**, and assigns a grade based on the percentage. The project demonstrates the practical use of basic web development concepts and JavaScript decision-making.

INPUT CODE

```
<!DOCTYPE html>
<html>

<head>
   <title>Grade Calculator</title>
    <style>
      body {
         font-family: Arial;
         background-color: lightblue;
         text-align: center;
      }
      .calculator {
         width: 400px;
         margin: 50px auto;
         padding: 25px;
         background-color: white;
         border-radius: 10px;
      }
      h2 {
         color: darkblue;
      }
      input {
         width: 90%;
```

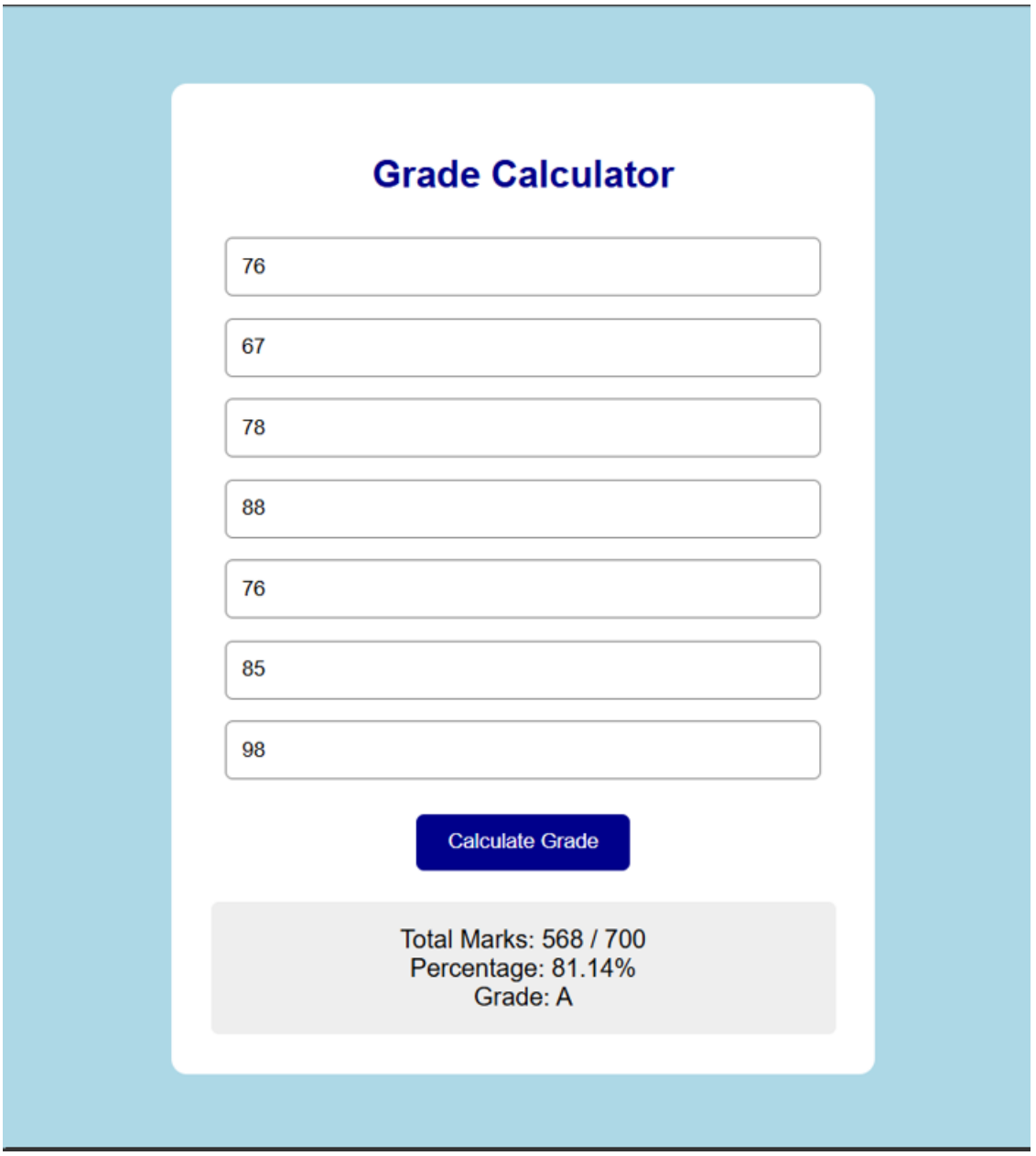
```
            padding: 10px;
            margin: 7px;
            border: 1px solid gray;
            border-radius: 5px;
        }
        button {
            padding: 10px 20px;
            margin-top: 15px;
            background-color: darkblue;
            color: white;
            border: none;
            border-radius: 5px;
            cursor: pointer;
        }
        button:hover {
            background-color: blue;
        }
        #result {
            margin-top: 20px;
            padding: 15px;
            background-color: #eeeeee;
            border-radius: 5px;
        }
    </style>
</head>
```

```
<body>
    <div class="calculator">
        <h2>Grade Calculator</h2>
        <input type="number" id="fft" placeholder="FFT Marks">
        <input type="number" id="ads" placeholder="ADS Marks">
        <input type="number" id="bad" placeholder="BAD Marks">
        <input type="number" id="pst" placeholder="PST Marks">
        <input type="number" id="cls" placeholder="CLS Marks">
        <input type="number" id="aui" placeholder="AUI Marks">
        <input type="number" id="dst" placeholder="DST Marks">
        <br>
        <button onclick="calculateGrade()">Calculate
Grade</button>
        <div id="result"></div>
    </div>
    <script>
        function calculateGrade() {
            let fft = document.getElementById("fft").value;
            let ads = document.getElementById("ads").value;
            let bad = document.getElementById("bad").value;
            let pst = document.getElementById("pst").value;
            let cls = document.getElementById("cls").value;
            let aui = document.getElementById("aui").value;
            let dst = document.getElementById("dst").value;
            if (fft == "" || ads == "" || bad == "" ||
                pst == "" || cls == "" || aui == "" || dst == "") {
```

```
            document.getElementById("result").innerHTML
="Please enter marks for all subjects.";
             return;
        }
        fft = Number(fft);
        ads = Number(ads);
        bad = Number(bad);
        pst = Number(pst);
        cls = Number(cls);
        aui = Number(aui);
        dst = Number(dst);
        if (fft < 0 || fft > 100 ||
            ads < 0 || ads > 100 ||
            bad < 0 || bad > 100 ||
            pst < 0 || pst > 100 ||
            cls < 0 || cls > 100 ||
            aui < 0 || aui > 100 ||
            dst < 0 || dst > 100) {
            document.getElementById("result").innerHTML =
                "Marks should be between 0 and 100.";
            return;
        }
        let total = fft + ads + bad + pst + cls + aui + dst;
        let percentage = total / 7;
        let grade;
        if (percentage >= 90) {
```

```
            grade = "A+";
        }
        else if (percentage >= 80) {
            grade = "A";
        }
        else if (percentage >= 70) {
            grade = "B";
        }
        else if (percentage >= 60) {
            grade = "C";
        }
        else if (percentage >= 40) {
            grade = "D";
        }
        else {
            grade = "F (Fail)";
        }
        document.getElementById("result").innerHTML =
            "Total Marks: " + total + " / 700<br>" +
            "Percentage: " + percentage.toFixed(2) + "%<br>" +
            "Grade: " + grade;
    }
  </script>
</body>
</html>
```

OUTPUT


Grade Calculator
76
67
78
88
76
85
98
Calculate Grade
Total Marks: 568 / 700
Percentage: 81.14%
Grade: A

CONCLUSION

The Grade Calculator provides a simple and efficient way to calculate a student's result. It checks whether marks are entered correctly and ensures that each mark is between **0 and 100**. JavaScript is used to calculate the total and percentage and determine the appropriate grade from **A+ to F**. Overall, this project helped demonstrate the use of **HTML for structure, CSS for styling, and JavaScript for calculations, validation, and logic**.